

# Delivery Package Index

VBF Case t2\_r3 | VBF-T2R3-IDX-01 | Virtual simulation calibre | Generated 2026-08-26

## Delivery Package Index — VBF Case t2\_r3

**Case name**: t2\_r3 — Grid energy storage cell (ED  $\geq 327.18$  Wh/kg · 4C no plating · SEI  $\leq 500/550$  nm ·  $-20$  °C  $\geq 90$  %)

**Numbering scheme**: `VBF-<CASE-ID-UPPERCASE>-<DOC-CODE>-<SEQ-NO>` (case id `t2\_r3` → `T2R3`)

**Generation date**: 2026-08-26

**Prepared**: \_\_\_\_\_ **Reviewed**: \_\_\_\_\_ **Approved**: \_\_\_\_\_

### Document codes

| Document code | Meaning                             | Corresponding file                                                                                           |
|---------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------|
| DS            | Specification                       | design_spec.md                                                                                               |
| BOM           | Bill of Materials                   | bom.xlsx                                                                                                     |
| DSH           | Datasheet technical parameter sheet | datasheet.docx                                                                                               |
| CALC          | Calculation sheet                   | calc.xlsx                                                                                                    |
| DVPR          | Design verification report          | dvpr.md                                                                                                      |
| DFMEA         | Failure analysis                    | dfmea.md                                                                                                     |
| IDX           | Delivery index (this file)          | delivery_index.md                                                                                            |
| CAD           | Structure model                     | <b>**not generated**</b> (optional deliverable; not requested in this headless case — no user clarification) |

### File list

| File                      | Number            | Format | Source description                                                                                                                                                                                            |
|---------------------------|-------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| design_spec.md            | VBF-T2R3-DS-01    | md     | cell design specification, generated per deliverable-design-spec spec; values from `cell/r7_final_*_dfn.json`, `r6_combo-v5-final_aging*_spme.json`, `r6_combo-v5-final_params.json`, Chen2020 parameter dump |
| design_spec.pdf           | VBF-T2R3-DS-01    | pdf    | PDF release of design_spec.md (reportlab)                                                                                                                                                                     |
| report.html               | VBF-T2R3-DS-02    | html   | seven-section activity report, mechanically rendered by `bda render` from log.jsonl (case root)                                                                                                               |
| bom.xlsx                  | VBF-T2R3-BOM-01   | xlsx   | bill of materials, dual calibre g/cell + kg/kWh (openpyxl); masses from calc-energy layer_kg_m2 × area; binder/CA split 96/2/2 = industry-typical estimate, annotated                                         |
| bom.pdf                   | VBF-T2R3-BOM-01   | pdf    | PDF release of bom.xlsx (reportlab)                                                                                                                                                                           |
| datasheet.docx            | VBF-T2R3-DSH-01   | docx   | customer-facing technical datasheet (python-docx); every field sourced from simulation outputs                                                                                                                |
| datasheet_docx_content.md | VBF-T2R3-DSH-02   | md     | content mirror of datasheet.docx used as the reportlab input for the PDF release (build intermediate)                                                                                                         |
| datasheet.pdf             | VBF-T2R3-DSH-01   | pdf    | PDF release of datasheet.docx (reportlab)                                                                                                                                                                     |
| calc.xlsx                 | VBF-T2R3-CALC-01  | xlsx   | design calculation sheet, 5 sheets: inputs → capacity/energy → energy density → N/P + mass → process parameters (openpyxl)                                                                                    |
| calc.pdf                  | VBF-T2R3-CALC-01  | pdf    | PDF release of calc.xlsx (reportlab)                                                                                                                                                                          |
| dvpr.md                   | VBF-T2R3-DVPR-01  | md     | design verification plan and report, virtual-test version; 5/5 contract criteria PASS at DFN calibre; N/A items listed                                                                                        |
| dvpr.pdf                  | VBF-T2R3-DVPR-01  | pdf    | PDF release of dvpr.md (reportlab)                                                                                                                                                                            |
| dfmea.md                  | VBF-T2R3-DFMEA-01 | md     | design FMEA, qualitative, simulation-signal based; highest risk = aged-cell 4C plating margin                                                                                                                 |
| dfmea.pdf                 | VBF-T2R3-DFMEA-01 | pdf    | PDF release of dfmea.md (reportlab)                                                                                                                                                                           |
| delivery_index.md         | VBF-T2R3-IDX-01   | md     | this delivery index (cover + controlled file list)                                                                                                                                                            |

|                    |                 |     |                                              |
|--------------------|-----------------|-----|----------------------------------------------|
| delivery_index.pdf | VBF-T2R3-IDX-01 | pdf | PDF release of delivery_index.md (reportlab) |
|--------------------|-----------------|-----|----------------------------------------------|

## Notes

- All concluded numerical values are mechanically sourced from tool outputs (PyBaMM DFN/SPMe via ``bda run-pyamm``, ``bda calc-energy``, ``bda log-evaluate`` round 7) — no values written from memory. Estimates (literature-transport overrides  $\sigma/D/t$ , SEI bridge  $k/V$ , binder/CA split, electrolyte density 1.2 g/cm<sup>3</sup>) are flagged inline.
- True DFT/MD endorsement skipped by case configuration (``real_compute = false``); recorded in the ``endorse`` log entry.
- Engineering manufacturability drawings (with tolerances), material specifications and line process cards are outside the pure-simulation boundary — not provided.
- Editable sources (md/docx/xlsx) retained alongside PDF releases; official deliverables are the PDF release versions.